MCIT 5920 Summer 2025

Homework 1 Questions

This assignment is due about one week from when it opens. The exact deadline and full instructions for submission are provided on Canvas. To receive full credit, your answers should be carefully justified. Each solution must be written independently by yourself - **no collaboration is allowed**.

Note. There are 5 homework problems for 50 points total immediately below. Before you work on them we recommend that you study the SOLVED problems that are at the end of this document. Please read carefully the solutions of these problems. They will help you formulate your own solutions to the homework problems immediately below.

Note. Whenever you use the Multiplication Rule (MR) or the Addition Rule (AR) in your solutions you have to state this explicitly: e.g., "by the multiplication (or addition) rule or by MR (or AR)", otherwise **1 point** will be deducted in **each** questions where this is applicable.

Note. Your assignment **must** be submitted as a LaTeX-formatted PDF, and you **must** use the provided ".tex" template. Submissions that are not in this format will not receive credit. In addition, please match the pages to the homework questions in Gradescope. This makes it much easier for us to grade your homework accurately and in a timely fashion. Submissions that are not page matched will be subject to a **2 point** deduction.

Homework Problems

(TO BE SOLVED AND SUBMITTED USING THE LATEX TEMPLATE PROVIDED).

1. [**10 pts**] Aliens are planning to attack $n \geq 3$ cities using the following devious methods: (1) installing fake cell towers, (2) deploying bird-like spy drones, (3) dropping frogs in the parks, and (4) conducting UFO fly-by's. The aliens have already decided that City A will be targeted with the first two methods, while Cities B and C will be subjected to exactly three of the four methods. For the remaining cities, each city may be attacked by any combination of these methods or be completely spared.

How many distinct attack plans can the aliens devise for these n cities?

2. [**10 pts**] For any two integers $m, n \in \mathbb{Z}$ prove that

(a) If m is odd and n is odd then $m - 2^n$ is odd.

(b) If m is even and n is odd then $(m+1)^2 n - m(n-1)^2$ is odd.

You can (but do not necessarily have to) use the rules of thumb in lecture and those for multiplication proved in Solved Problem below.

3. [**10 pts**] The city of Ploieşti has $3n$ recycling points where $n \geq 6$. The recycling points are distinguishable and numbered from 1 to $3n$. Hoping to get reelected, but having a shortage of recycling bins Mayor Val decides to place:

(a) At each recycling point numbered from 1 to $n-3$ a bin for plastic, and one for paper.

(b) At each recycling point numbered from $n-2$ to $n+3$ a bin for glass, *or* one for paper.

(c) At each recycling point numbered from $n+4$ to $2n-1$ a bin for plastic, *or* one for glass *or* one for paper.

(d) At each recycling point numbered from $2n$ to $3n$ a sign that says "Vote Val (the Best Recycler) for Mayor".

How many distinct recycling landscapes can Ploieşti possibly have?

4. [**10 pts**] Prove that for all integers x, y, if x and y are both odd, then $x^2 - y^2$ is divisible by 8.

5. [**10 pts**] Calculate the following parts and justify your answer.

(a) Let $A_0 = \emptyset$, $A_1 = A_0 \cup \{\{A_0\}\}$, $A_2 = A_1 \cup \{A_1\}$, $A_3 = A_2 \cup \{A_2\}$. Calculate $|A_0 \cup A_1 \cup A_2 \cup A_3|$. Justify your answer.

(b) Let $Q_0 = \{7\}$, $Q_1 = Q_0 \times \{Q_0\}$, $Q_2 = Q_1 \times \{\{Q_1\}\}$, $Q_3 = Q_2 \times \{Q_2\}$. Calculate $|Q_0 \cup Q_1 \cup Q_2 \cup Q_3|$. Justify your answer.

Solved Problems

A. Consider the name VAL TANNENU. Looking up the letters of this name on an US phone keypad we read the following numbers

825 8266368

or, written as an US phone number (825) 826-6368. So we have a correspondence between phone numbers and "people" whose first name has three letters and last name has seven letters. (We consider only capital letters.)

(a) Now consider the US phone number (484) $898 - 2665$. To how many different names does this number correspond?

(b) What is the maximum number of distinct letters that can appear in a name that corresponds to an US phone number? (The minimum is obviously 1, for a number like $(222) - 222 - 2222$. The maximum will be for another kind of number.) Just state the answer and explain how you obtained it. You do not need to *prove* that it is the maximum.

Answer

(a) To construct a name given the phone number provided, we select a letter for each digit based on the US keypad. The digit 4 corresponds to the letters GHI, and 8 the letters TUV. For the first letter select a G, H, or I, so by addition rule there are 3 choices. For the second letter select a T, U, or V, which is 3 choices by the addition rule, and 3 choices for the third letter, selecting G, H,or I again. Since there are 3 ways to perform each of these 3 steps, by the multiplication rule there are a total of 27 possibilities for the first name.

9 corresponds to WXYZ, 2 to ABC, 6 to MNO, and 5 to JKL. Thus for the code 8982665, we can construct a name, in steps, as follows:

Select a letter for the number 8 - there are 3 ways to do this (JHI)

Select a letter for 9: 4 ways (WXYZ)

Select a letter for 8: 3 ways (JHI)

Select letter for 2: 3 ways (ABC)

Select for 6 twice (3 ways to select letter for first 6 and 3 ways to select for second 6, by multiplication rule this makes 9 ways)

Select letter for 5: 3 ways (JKL)

By multiplication rule there are $3*4*3*3*9*3 = 2,916$ ways to construct the last name

By the multiplication rule it follows that there are $27 * 2,916 = 78,732$ total names that can be constructed by this number

(b) The maximum number of distinct letters that can appear in a name corresponding to a US phone number is 10

A US phone number corresponding to a name consists of 10 digits, which maps to a name with 3 letters in the first name and 7 letters in the last name. Since we need to assign one letter per digit to form the name, and there are only 10 digits, the maximum number of distinct letters we can have in the name is 10.

B. For each of the following 3 statements, give examples of finite sets A and B that make the statement TRUE. You do not have to prove that the statement is true, just write down the examples.

(a) $2^A \cap 2^B = 2^{A \cap B}$.

(b) $2^A \setminus 2^B \subseteq 2^{A \setminus B}$.

(c) $A \times B \subseteq (A \setminus B) \times (B \setminus A)$

Answer

(a) $A = \{\}$ and $B = \{1\}$. Then, $2^A = \{\{\}\}$ and $2^B = \{\{\}, \{1\}\}$, so $2^A \cap 2^B = \{\{\}\}$. Also, $A \cap B = \{\}$, so $2^{A \cap B} = \{\{\}\}$.

(b) $A = \{1\}$ and $B = \{2\}$. Then, $2^A = \{\{\}, \{1\}\}$ and $2^B = \{\{\}, \{2\}\}$, so $2^A \setminus 2^B = \{\{1\}\}$. Also, $A \setminus B = \{1\}$, so $2^{A \setminus B} = \{\{\}, \{1\}\}$. Hence, every element in the LHS is also in the RHS.

(c) $A = \{1\}$ and $B = \{2\}$. Then, $A \times B = \{(1, 2)\}$. Also, $A \setminus B = \{1\}$ and $B \setminus A = \{2\}$ so $(A \setminus B) \times (B \setminus A) = \{(1, 2)\}$. Hence, every element in the LHS is also in the RHS.

C. Let n be a positive integer. $2n$ distinguishable Hogwarts students, n from Gryffindor and n from Slytherin participate in Professor Snape's experiment. Snape has at his disposal potion A, potion B, and potion C. Each student from Gryffindor is given one of the following concoctions: potion A, or potion B, or a mixture of A and B. Each student from Slytherin is given one of the following concoctions: a mixture of A and B, or a mixture of B and C.

(a) In how many distinct ways could Snape have distributed his concoctions?

(b) Same question if, embarrassingly, Snape discovers that potion A is actually the same as potion B.

Answer

(a) First we use *the Addition rule* to count the number of concoctions since these concoctions are distinct to each other. For Gryffindor students, there are 3 different concoctions: A, B, AB. For Slytherin students, there are 2 different concoctions: AB, BC.

Now we count the number of ways to distribute the concoctions to the **Gryffindor** students in n steps (each step corresponds to one student). Since each step is independent of each other, we can use *the Multiplication rule.*

- Step 1: Assign a concoction to the first student. There are 3 ways to do this.
- Step 2: If $n \geq 2$, Assign a concoction to the second student. There are 3 ways to do this.
- Step 3: ...
- Step n: If there are more than $n - 2$ students, assign a concoction to the nth student. There are 3 ways to do this.

Therefore the total number of ways to distribute the concoctions are: 3^n

Now we count the number of ways to distribute the concoctions to the **Slytherin** students in n steps (each step corresponds to one student). Since each step is independent of each other, we can use *the Multiplication rule.*

- Step 1: Assign a concoction to the first student. There are 2 ways to do this.
- Step 2: If $n \geq 2$, Assign a concoction to the second student. There are 2 ways to do this.
- Step 3: ...
- Step n: If there are more than $n-2$ students, assign a concoction to the nth student. There are 2 ways to do this.

Therefore the total number of ways to distribute the concoctions are: 2^n

Overall, since assigning concoctions to the Gryffindor and to the Slytherin students occur in two independent steps, by the Multiplication Rule, we get a total of $3^n 2^n$ ways for Snape to distribute his concoctions.

(b) We will use A to refer to the concoctions A, B, and AB, which are now identical. First we use *the Addition rule* to count the number of concoctions since these concoctions are distinct to each other. For Gryffindor students, there is only 1 concoction, namely A, since (A, B, and AB are all the same). For Slytherin students, there are 2 different concoctions: A (previously AB) and AC (previously BC).

Now we count the number of ways to distribute the concoctions to the **Gryffindor** students in n steps (each step corresponds to one student). Since each step is independent of each other, we can use *the Multiplication rule.* Each of the n students has 1 concoction option, so overall there are $1^n = 1$ way to make the distribution.

Since for **Slytherin** there are 2 different concoctions, the total number of ways to distribute the concoctions is 2^n, similarly to part (a).

Overall, since assigning concoctions to the Gryffindor and to the Slytherin students occur in two independent steps, by the Multiplication Rule, we get a total of $1 * 2^n = 2^n$ ways for Snape to distribute his concoctions.

D. Let m and n be two integers such that $m < n$ and such that m and n have different parity. Define

$$A_{m,n} = \{z \in \mathbb{Z} \mid m < z < n\}$$

Prove that if $A_{m,n} \neq \emptyset$ then $|A_{m,n}| \geq 2$.

Answer

Let m and n be two integers such that $m < n$ and such that m and n have different parity.

Assume that $A_{m,n} \neq \emptyset$. Since m and n have different parity we have two cases.

Case 1 m is even and n is odd.

Consider $z_1 = m + 1$ and $z_2 = m + 2$. Since m is even, z_1 is odd and z_2 is even. We cannot have $n = z_1$ because then $A_{m,n} \neq \emptyset$. Also, we cannot have $n = z_2$ because n is odd and z_2 is even. It follows that $n > z_2$. Then, both z_1 and z_2 are elements of $A_{m,n}$. Therefore $|A_{m,n}| \geq 2$.

Case 2 m is odd and n is even.

Consider $z_1 = m+1$ and $z_2 = m+2$. Since m is odd, z_1 is even and z_2 is odd. We cannot have $n = z_1$ because then $A_{m,n} \neq \emptyset$. Also, we cannot have $n = z_2$ because n is even and z_2 is odd. It follows that $n > z_2$. Then, both z_1 and z_2 are elements of $A_{m,n}$. Therefore $|A_{m,n}| \geq 2$.

Important remark Case 2 is like Case 1 except that "even" and "odd" are swapped! Such situations happen often in mathematical proofs and we don't want to essentially repeat ourselves unnecessarily! The custom is to to write only the following:

Case 2 is similar (or analogous)

We will accept writing what you just saw in Case 2 when we grade your solutions, but only if it is indeed similar or analogous! If you are not sure, it's better to write the solution in full.

Alternative Solution

$A_{m,n}$ is the set containing all the integers comprised between m and n, exclusive. It follows that the smallest element in this set is $m + 1$ and the largest element in this set is $n - 1$.

We assume $A_{m,n} \neq \emptyset$. This means $A_{m,n}$ contains at least one element.

Let's assume towards a contradiction that $A_{m,n}$ contains exactly one element. Then, we must have $m + 1 = n - 1$

$m + 1 = n - 1 \iff m + 2 = n$. But we know that m and n have different parities. Without loss of generality, let m be even and n be odd. As the sum of two even numbers, $m+2$ is even. Since n is odd, n can't be equal to $m + 2$. We have reached a contradiction. We can conclude that if $A_{m,n} \neq \emptyset$, then $A_{m,n}$ contains at least two distinct elements: $m + 1$ and $n - 1$.

The case where m is odd and n is even yields similar results.

E. You have seen in Module 1, segment "Even and Odd", the so-called rules of thumb for parity of the *sum* and the *difference* of two integers. You can also prove these rules of thumb and in this problem we ask you to prove one of the rules of thumb for the sum and one of them for difference and then the rules of thumb for multiplication. By the way, you can use any of these rules in any homework or exam problem without proving them.

For any two integers $m, n \in \mathbb{Z}$ prove that

(a) If m is odd and n is even then $m + n$ is odd.

(b) If m is odd and n is odd then $m - n$ is even.

(c) If m is even and n is even then mn is even.

(d) If m is odd and n is odd then mn is odd.

(e) If m is even and n is odd then mn is even.

Answer

(a) If m is odd then $m = 2k + 1$ for some $k \in \mathbb{Z}$.

If n is even then $n = 2l$ for some $l \in \mathbb{Z}$.

Thus $m + n = 2k + 1 + 2l = 2(k + l) + 1$, so $m + n = 2p + 1$ for some $p \in \mathbb{Z}$, and so $m + n$ is odd.

(b) If m is odd then $m = 2k + 1$ for some $k \in \mathbb{Z}$.

If n is odd then $n = 2l + 1$ for some $l \in \mathbb{Z}$.

Thus $m - n = 2k + 1 - 2l - 1 = 2(k - l)$, so $m - n = 2p$ for some $p \in \mathbb{Z}$, and so $m - n$ is even.

(c) If m is even then $m = 2k$ for some $k \in \mathbb{Z}$.

If n is even then $n = 2l$ for some $l \in \mathbb{Z}$.

Thus $mn = (2k)(2l) = 2(2kl)$, so $mn = 2p$ for some $p \in \mathbb{Z}$, and so mn is even.

(d) If m is odd then $m = 2k + 1$ for some $k \in \mathbb{Z}$.

If n is odd then $n = 2l + 1$ for some $l \in \mathbb{Z}$.

Thus $mn = (2k + 1)(2l + 1) = 4kl + 2k + 2l + 1 = 2(2kl + k + l) + 1$, so $mn = 2p + 1$ for some $p \in \mathbb{Z}$, and so mn is odd.

(e) If m is even then $m = 2k$ for some $k \in \mathbb{Z}$.

If n is odd then $n = 2l + 1$ for some $l \in \mathbb{Z}$.

Thus $mn = (2k)(2l + 1) = 2(2kl + k)$, so $mn = 2p$ for some $p \in \mathbb{Z}$, and so mn is even.